# MILITARY SPECIFICATION

## MODULES, STANDARD ELECTRONIC

## DRIVER, LAMP/RELAY, DIGITAL, KEY CODE FQD

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic digital lamp/relay driver module.

* 1.2 Classification. This specification covers two classes of modules.

Class I and class II as specified in MIL-STD-1389.

## 2. APPLICABLE DOCUMENTS

* 2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

## 3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.



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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.527 failures per million operating hours for class I or 0.795 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, failure rate for class I shall be used for reliability prediction (see 6.3).

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I. Class I modules shall conform to the electrical requirements of class I only, and class II modules shall conform to the electrical requirements of class I and class II (see 6.3).

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be FQD.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/278-1 for class I and M28787/278-2 for class II, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage ($V_{CC}$) of 5.0 ±0.5 V dc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 V dc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

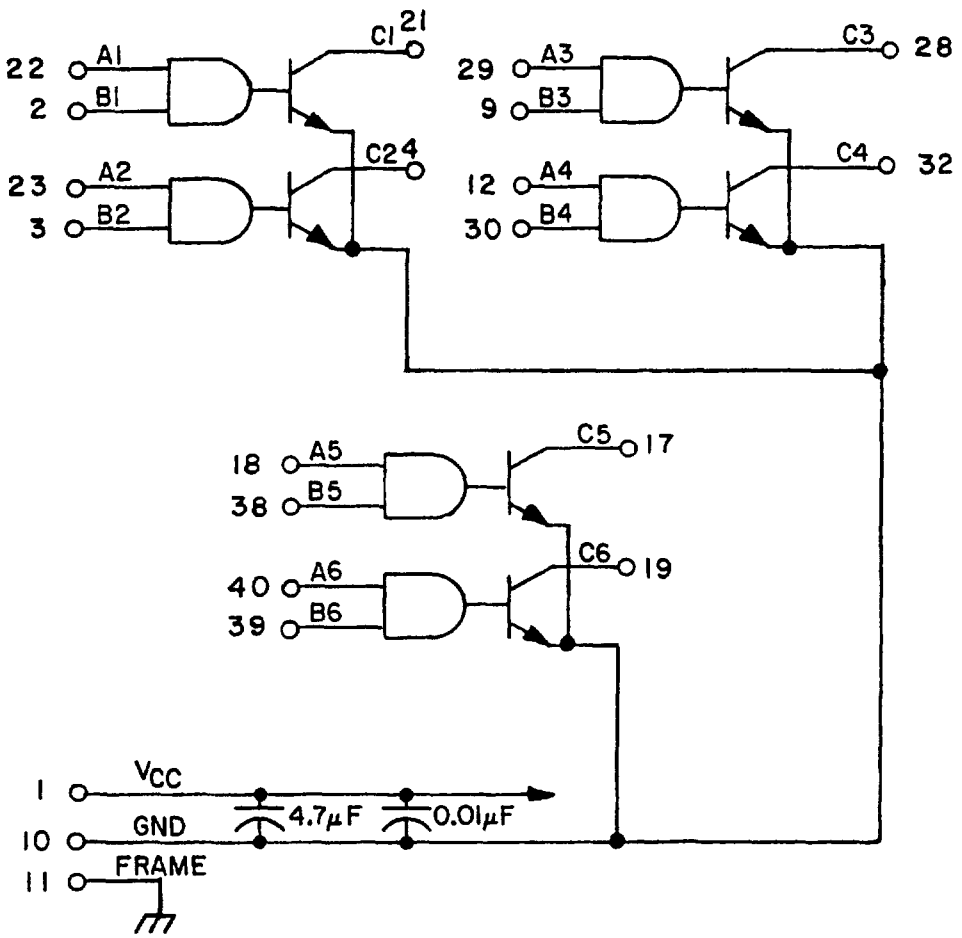
4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

* 4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

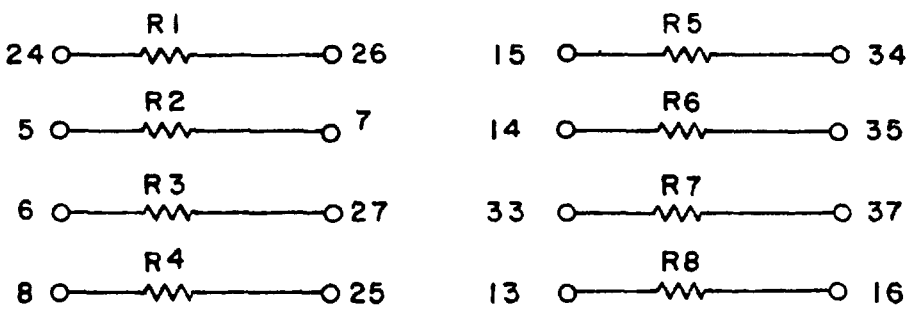
4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

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R1 - R8 = 1kΩ ± 5 %, 1/4 W




UNUSED PINS. 20, 31, AND 36

* FIGURE 1. FQD module function diagram, lamp/relay driver.



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*TABLE I. Module electrical requirements.

| Test condition | Parameter | Ref test table | Test type 1/-3/ | 25°C ±5°C 4/ | | Class I 4/ | | | | Class II 4/ | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | 0°C ±5°C | | 60°C ±5°C | | -55°C ±5°C | | 85°C ±5°C | | |
| | | | | Min | Max | Min | Max | Min | Max | Min | Max | Min | Max | |
| A1 | Continuity | III | X | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | Ω |
| A2-A4 | Isolation | | | | | | | | | | | | | |
| | Initial | III | S | 10 | ∞ | 10 | ∞ | 10 | ∞ | 10 | ∞ | 10 | ∞ | MΩ |
| | End-of-life | III | S | 1 | ∞ | 1 | ∞ | 1 | ∞ | 1 | ∞ | 1 | ∞ | MΩ |
| B1-B12 | Input breakdown current | IV | S | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | mA |
| C1-C12 | Input clamp diode voltage | V | X | -0.4 | -1.5 | -0.4 | -1.5 | -0.4 | -1.5 | -0.4 | -1.5 | -0.4 | -1.5 | V |
| D1-D12 | Output leakage current | VI | X | 0 | 300 | 0 | 300 | 0 | 300 | 0 | 300 | 0 | 300 | µA |
| E1-E6 | Output voltage (low level), $I_{OL}$ = 300 mA | VII | X | 0 | 800 | 0 | 800 | 0 | 800 | 0 | 800 | 0 | 800 | mV |
| E7-E12 | Output voltage (low level), $I_{OL}$ = 100 mA | VII | X | 0 | 500 | 0 | 500 | 0 | 500 | 0 | 500 | 0 | 500 | mV |
| F1-F12 | Input current, high level | VIII | X | 0 | 40 | 0 | 40 | 0 | 40 | 0 | 40 | 0 | 40 | µA |
| G1-G12 | Input current, low level | IX | X | -0.4 | -1.73 | -0.4 | -1.73 | -0.4 | -1.73 | -0.4 | -1.73 | -0.4 | -1.73 | mA |
| H1-H8 | Resistance | X | X | 950 | 1,050 | 890 | 1,110 | 900 | 1,100 | 890 | 1,100 | 900 | 1,100 | Ω |
| J1 | Power supply current, low level outputs | XI | S | 1 | 213 | 1 | 213 | 1 | 213 | 1 | 213 | 1 | 213 | mA |
| J2 | Power supply current, high level outputs | XI | S | 1 | 42 | 1 | 42 | 1 | 42 | 1 | 42 | 1 | 42 | mA |
| K1-K6 | Transition time ($t_{THL}$), $C_L$ = 50 pF | XII | S | 2 | 14 | 2 | 14 | 2 | 15 | 2 | 14 | 2 | 16 | ns |
| L1-L12 | Propagation delay time ($t_{PLH}$), $C_L$ = 50 pF | XIII | S | 3 | 36 | 3 | 36 | 3 | 38 | 3 | 36 | 3 | 40 | ns |
| M1-M12 | Propagation delay time ($t_{PHL}$), $C_L$ = 50 pF | XIII | S | 3 | 36 | 3 | 36 | 3 | 38 | 3 | 36 | 3 | 40 | ns |
| N | Power supply transient amplitude | XIV | Q | 0 | 400 | 0 | 400 | 0 | 400 | 0 | 400 | 0 | 400 | mV |

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See footnotes at end of table.





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*TABLE I. Module electrical requirements - Continued.

| Test condition | Parameter | Ref test table | Test type 1/-3/ | 25°C ±5°C 4/ | | Class I 4/ | | | | Class II 4/ | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | 0°C ±5°C | | 60°C ±5°C | | -55°C ±5°C | | 85°C ±5°C | | |
| | | | | Min | Max | Min | Max | Min | Max | Min | Max | Min | Max | |
| P | Power supply filter capacitance | XV | S | 3.3 | 6.1 | 3.3 | 6.1 | 3.3 | 6.1 | 3.3 | 6.1 | 3.3 | 6.1 | µF |
| Q1-Q12 | Input capacitance | XVI | Q | 2 | 15 | 2 | 15 | 2 | 15 | 2 | 15 | 2 | 15 | pF |

1/ "X" parameters shall be 100 percent tested during module acceptance tests.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified, limits are for initial and end-of-life.



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4.4.2 Order of testing. Test conditions A through Q shall be performed in any order with the following exceptions. If test conditions A, B, or C are part of a sequence of tests, the order of testing shall be as follows: A, B, C where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through Q in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions C through P shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 4.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 5. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column and test condition P shall be performed using the initial limits at 25°C ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests C through P shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests C through Q shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in a class I environment, the failure rate for class I shall be used for reliability prediction (see 3.3 and 3.4).

* 6.4 Module description. The following electrical parameters reflect the 0°C to 60°C and -55°C to 85°C end-of-life limits from table I.

| | Class I | Class II | Unit |
|---|---|---|---|
| Supply voltage | 5 ±10 percent | 5 ±10 percent | V dc |
| Maximum power dissipation/ IC module | | | |
| Outputs low, loaded ($V_{CC}$ = 5.5 V) | 800 (derated 6.6 mW/°C above 29°C $T_A$) | 800 (derated 6.6 mW/°C above 29°C $T_A$) | mW |

NOTE: Both halves of dual-circuit modules may conduct rated current simultaneously, but average short-time power dissipation must not exceed this value.



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| | Class I | Class II | Unit |
|---|---|---|---|
| Outputs high, loaded ($V_{OH}$ = 30 V, $V_{CC}$ = 5.5 V) | 115 | 115 | mW |
| Maximum power dissipation/module | | | |
| Outputs low, loaded ($V_{OH}$ = 30 V, $V_{CC}$ = 5.5 V) | 2,400 | 2,400 | mW |
| Outputs high, loaded ($V_{OH}$ = 30 V, $V_{CC}$ = 5.5 V) | 345 | 345 | mW |
| Input voltage | | | |
| Low level | 0.8 maximum | 0.8 maximum | V |
| High level | 2 minimum, $V_{CC}$ maximum | 2 minimum, $V_{CC}$ maximum | V |
| Output voltage | | | |
| Low level ($I_{OL}$ = 100 mA) | 0.5 maximum | 0.5 maximum | V |
| ($I_{OL}$ = 300 mA, $V_{CC}$ = 4.5 V) | 0.8 maximum | 0.8 maximum | V |
| High level (applied to an output in the off-state) | 30 maximum | 30 maximum | V |
| Input current | | | |
| Low level ($V_{IL}$ = 0.4 V, $V_{CC}$ = 5.5 V) | -1.6 maximum | -1.6 maximum | mA |
| High level ($V_{IH}$ = 2.4 V, $V_{CC}$ = 5.5 V) | 40 maximum | 40 maximum | µA |
| Output current | | | |
| Low level ($I_{OL}$ @ 85°C) | 200 maximum | 200 maximum | mA |
| ($I_{OL}$ @ -55°C, 0°C, 25°C, and 60°C) | 300 maximum | 300 maximum | mA |
| High level ($V_{OH}$ = 30 V, $V_{CC}$ = 4.5 V) | 300 maximum | 300 maximum | µA |
| Power supply current, IC, total | | | |
| Outputs low, no load ($V_{CC}$ = 5.5 V) | 213 maximum | 213 maximum | mA |
| Outputs high, no load ($V_{CC}$ = 5.5 V) | 42 maximum | 42 maximum | mA |

* 6.5 Module function. This module contains six independent indentical lamp or relay driver circuits and eight pin-programmable resistors.

* 6.5.1 Input to output relationships. Each independent lamp/relay driver performs NAND logic. A 2-input AND gate is followed by a high-voltage open collector transistor output driver. The input-to-output relationship shall be as shown in table II.

TABLE II. Input to output relationship. 1/ 2/

| INPUTS | | OUTPUT |
|---|---|---|
| A | B | C |
| L | L | H |
| L | H | H |
| H | L | H |
| H | H | L |




1/ L = Low level.
2/ H = High level.



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* 6.5.2 Output consideration. Load power must be supplied externally to appropriate module circuit pins. The negative return of this supply must be made common with the negative return of the module ±5 V supply. Each lamp/relay driver is capable of sinking any load current less than 300 mA at -55°C, 0°C, 25°C, and 60°C with the output driver in the low-level (on) state. At 85°C, each driver is capable of sinking any current less than 200 mA or any one driver (on) and I.C. is capable of sinking a maximum of 300 mA with the other driver in the (off) state. Each lamp/relay driver is capable of withstanding any output voltage of less than 30 V dc with the output driver in the high-level (off) state. If inductive loads (relays) are used, inductive voltage spikes must be suppressed to limit the $V_{CE}$ voltage to less than 30 volts. If lamps are driven, external bleeder resistors may be required to limit cold lamp in-rush current to less than or equal to 600 mA. If any driver on the FQD module is driving a lamp/relay or any other inductive load, none of the remaining drivers shall be used with logic circuits.

6.5.3 Programmable resistors. Each module provides eight independently programmable, 1,000 ohms, 1/4 watt resistors, which may be used as pullup resistors for unused logic inputs. Care should be taken, however, that power dissipation limits are not exceeded in using these resistors.

6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

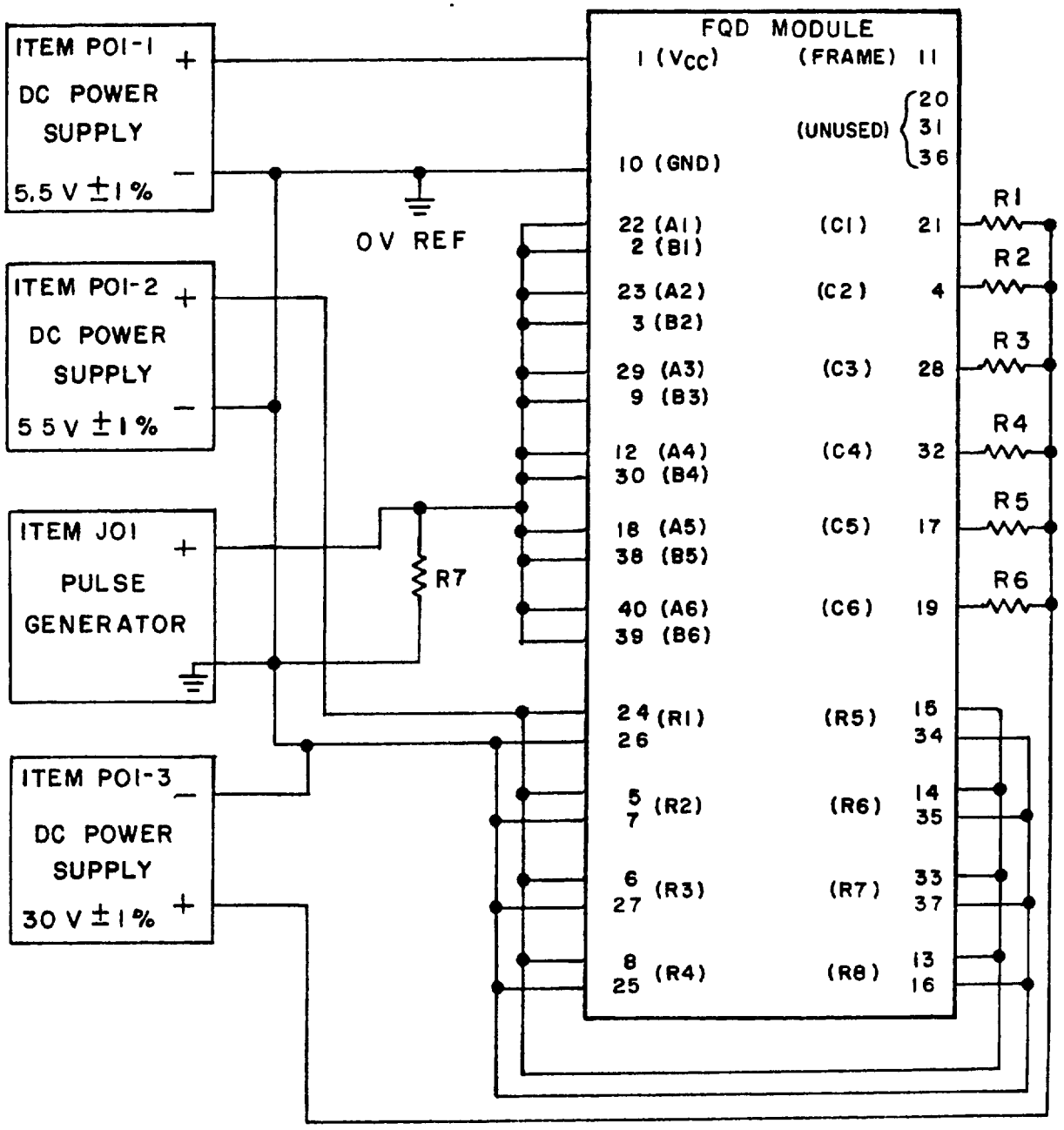
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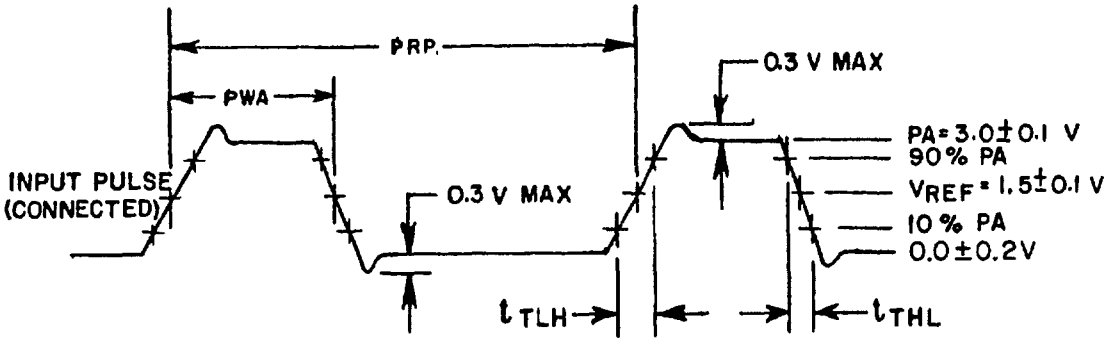



NOTES:
1. R1-R6 = 147Ω ±1%, 6 W.
2. R7 = 51Ω ±5%, 1/2 W.
3. Adjust item JO1 in accordance with figure 3.

FIGURE 2. Operating temperature and life test circuit.



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NOTE: Input pulse parameters:
$t_{TLH} = t_{THL} \leq 15$ ns.
PWA = 25 ±5 µs.
PRP = 50 ±5 µs.

FIGURE 3. Operating temperature and life test circuit waveform.



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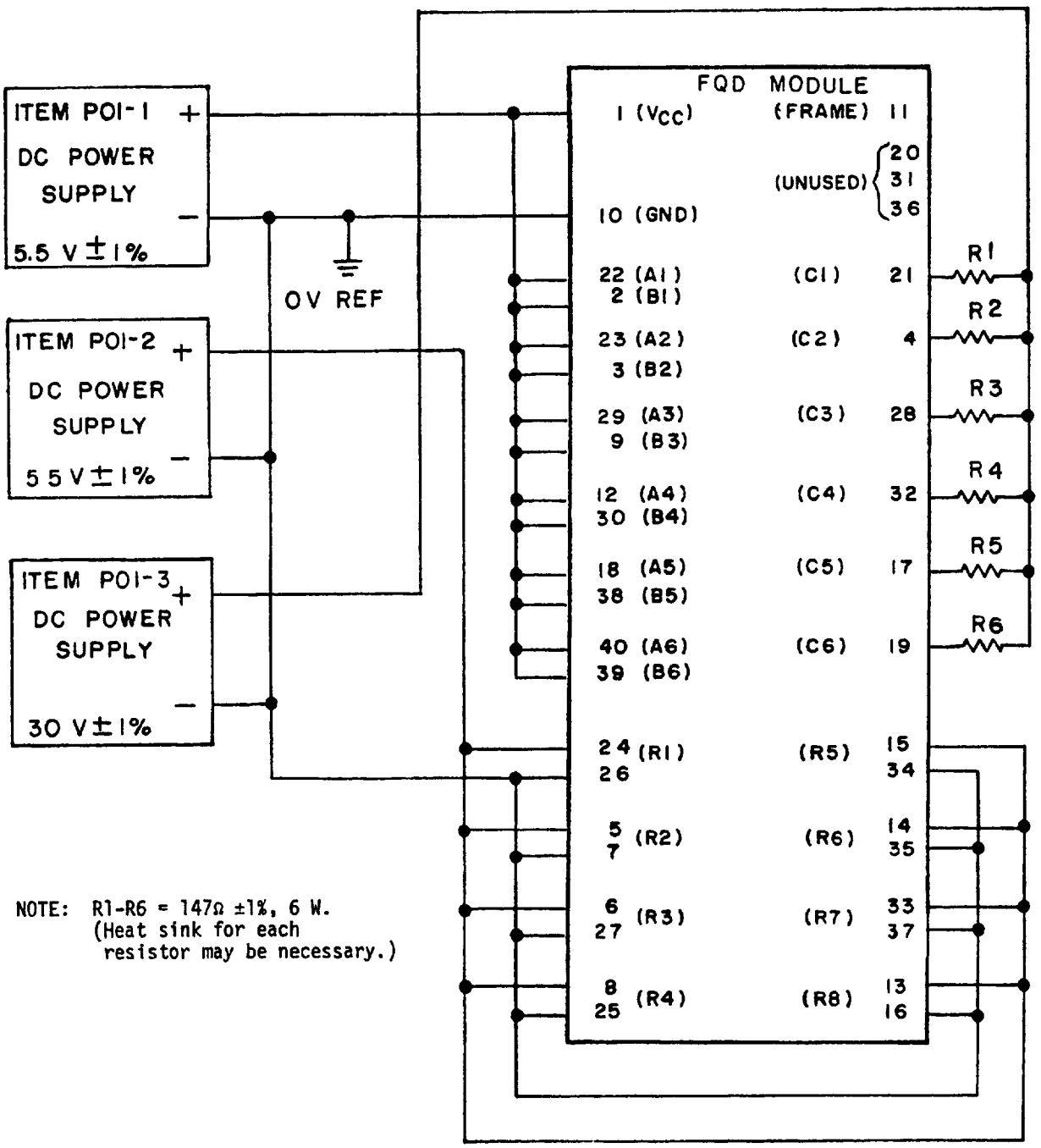



FIGURE 4. Component temperature test circuit.



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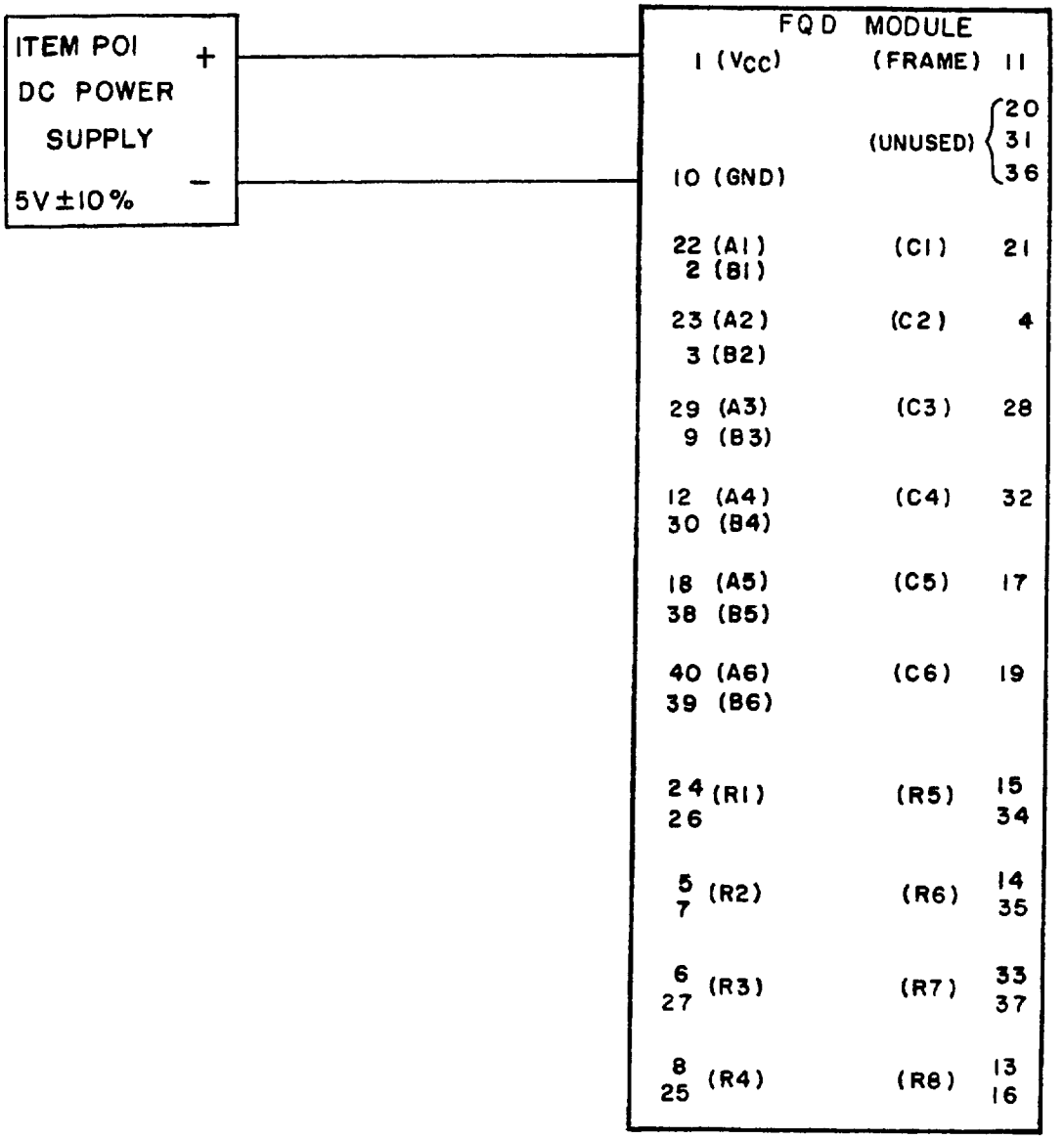



FIGURE 5. Vibration test circuit.

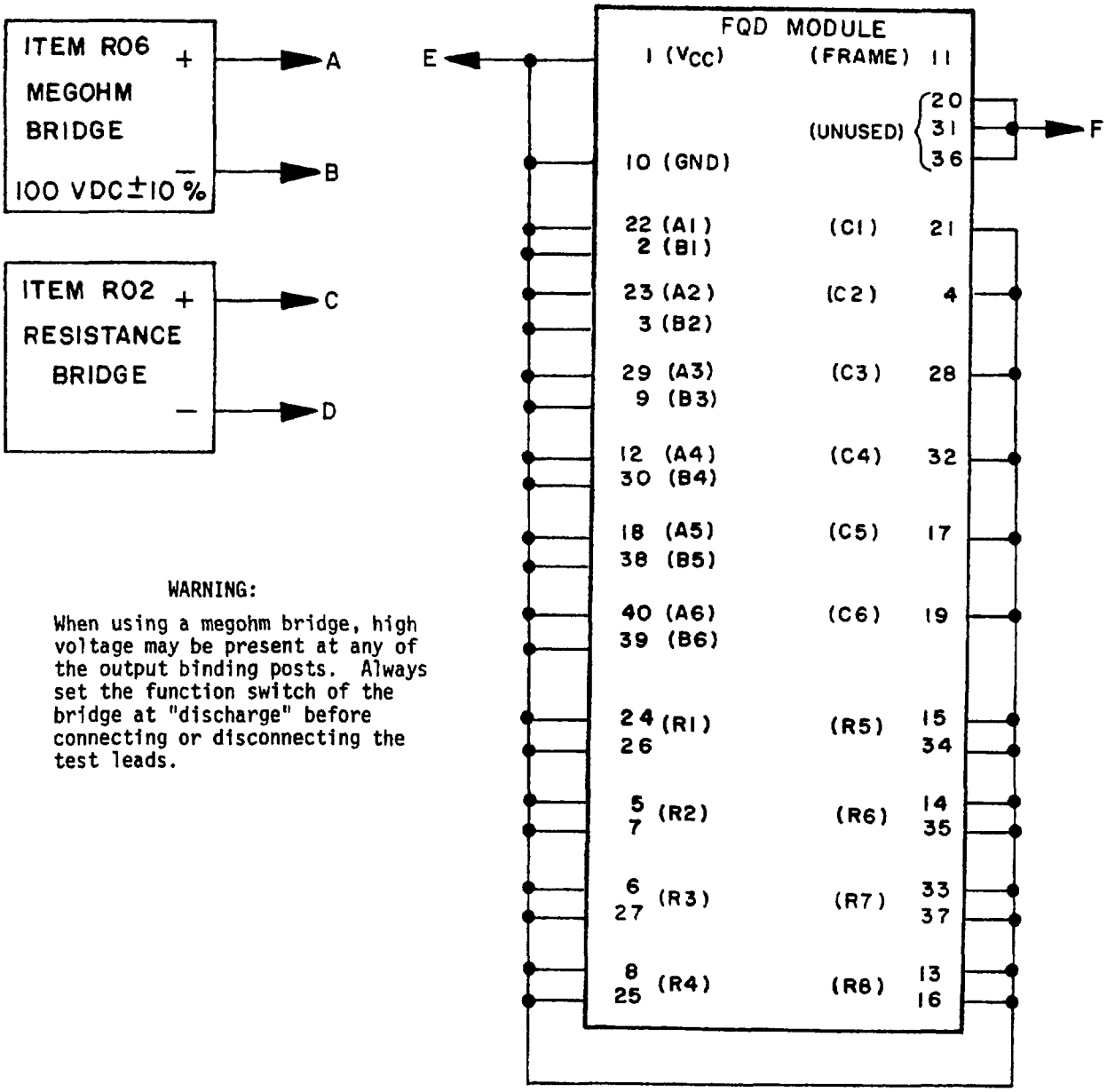
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*TABLE III. Continuity and isolation test.

| Test condition | Parameter | Connect | | | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | D to pin | Min | Max | | |
| A1 | Continuity | 11 | Frame | Open | Open | 0 | 1 | Ω | 6 |
| A2 | Isolation | Open | Open | E | 11 | 10 | ∞ | MΩ | " |
| A3 | " | " | " | E | F | " | " | " | " |
| A4 | " | " | " | F | 11 | " | " | " | " |



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WARNING:

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the function switch of the bridge at "discharge" before connecting or disconnecting the test leads.

NOTE: Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal for test condition A1.

* FIGURE 6. Continuity and isolation test circuit for table III.



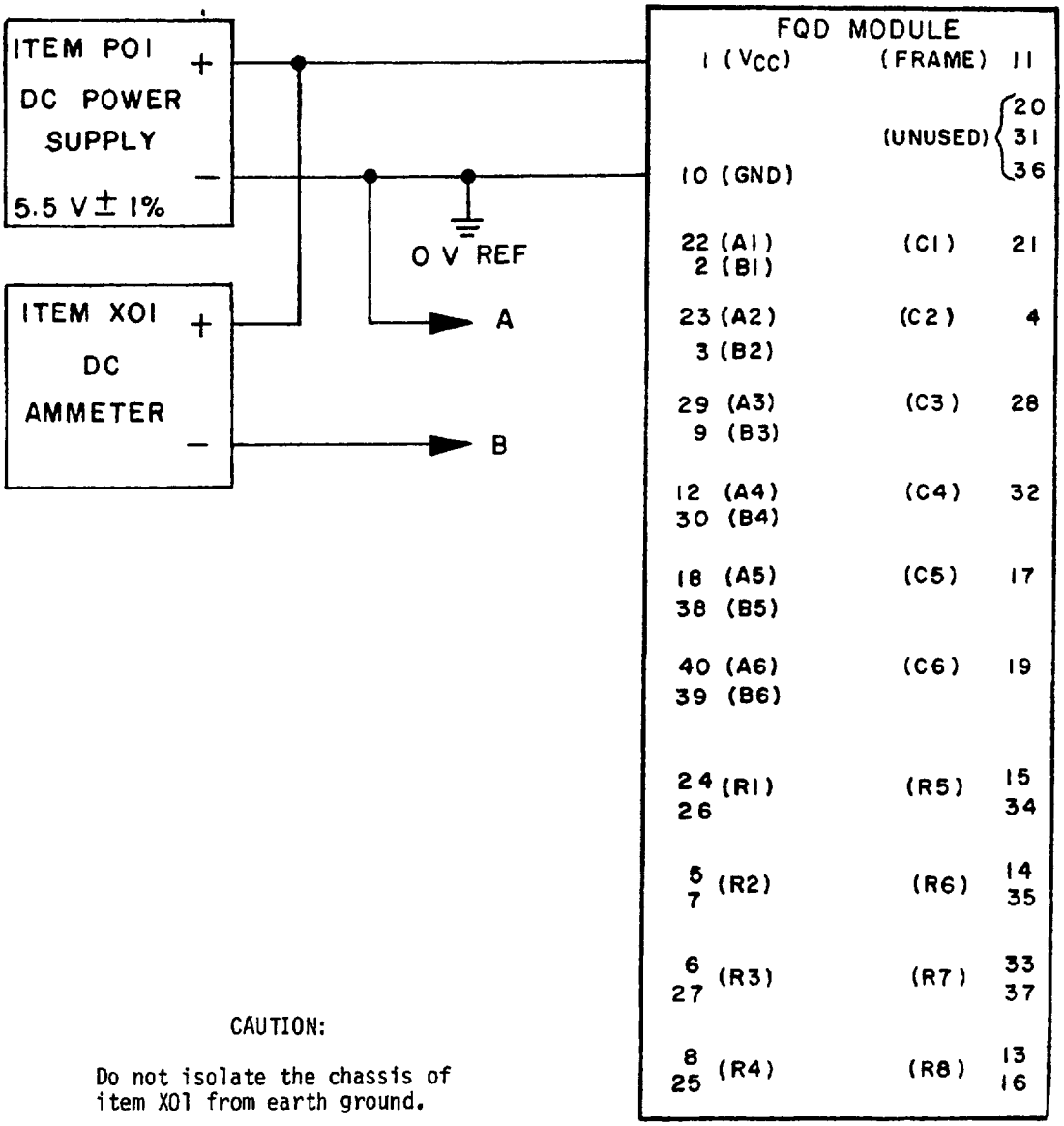
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TABLE IV. Input breakdown current test.

| Test condition | Parameter | | Connect | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|
| | | | A to pin | B to pin | Min | Max | | |
| B1 | A1 | Input breakdown current | 2,3,9,12,18,23, 29,30,38,39,40 | 22 | 0 | 1 | mA | 7 |
| B2 | B1 | " | 3,9,12,18,22, 23,29,30,38, 39,40 | 2 | " | " | " | " |
| B3 | A2 | " | 2,3,9,12,18,22, 29,30,38,39,40 | 23 | " | " | " | " |
| B4 | B2 | " | 2,9,12,18,22, 23,29,30,38,39, 40 | 3 | " | " | " | " |
| B5 | A3 | " | 2,3,9,12,18,22, 23,30,38,39,40 | 29 | " | " | " | " |
| B6 | B3 | " | 2,3,12,18,22, 23,29,30,38,39, 40 | 9 | " | " | " | " |
| B7 | A4 | " | 2,3,9,18,22,23, 29,30,38,39,40 | 12 | " | " | " | " |
| B8 | B4 | " | 2,3,9,12,18,22, 23,29,38,39,40 | 30 | " | " | " | " |
| B9 | A5 | " | 2,3,9,12,22,23, 29,30,38,39,40 | 18 | " | " | " | " |
| B10 | B5 | " | 2,3,9,12,18,22, 23,29,30,39,40 | 38 | " | " | " | " |
| B11 | A6 | " | 2,3,9,12,18,22, 23,29,30,38,40 | 39 | " | " | " | " |
| B12 | B6 | " | 2,3,9,12,18,22, 23,29,30,38,39 | 40 | " | " | " | " |



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CAUTION:

Do not isolate the chassis of item X01 from earth ground.

FIGURE 7. Input breakdown current test circuit for table IV.

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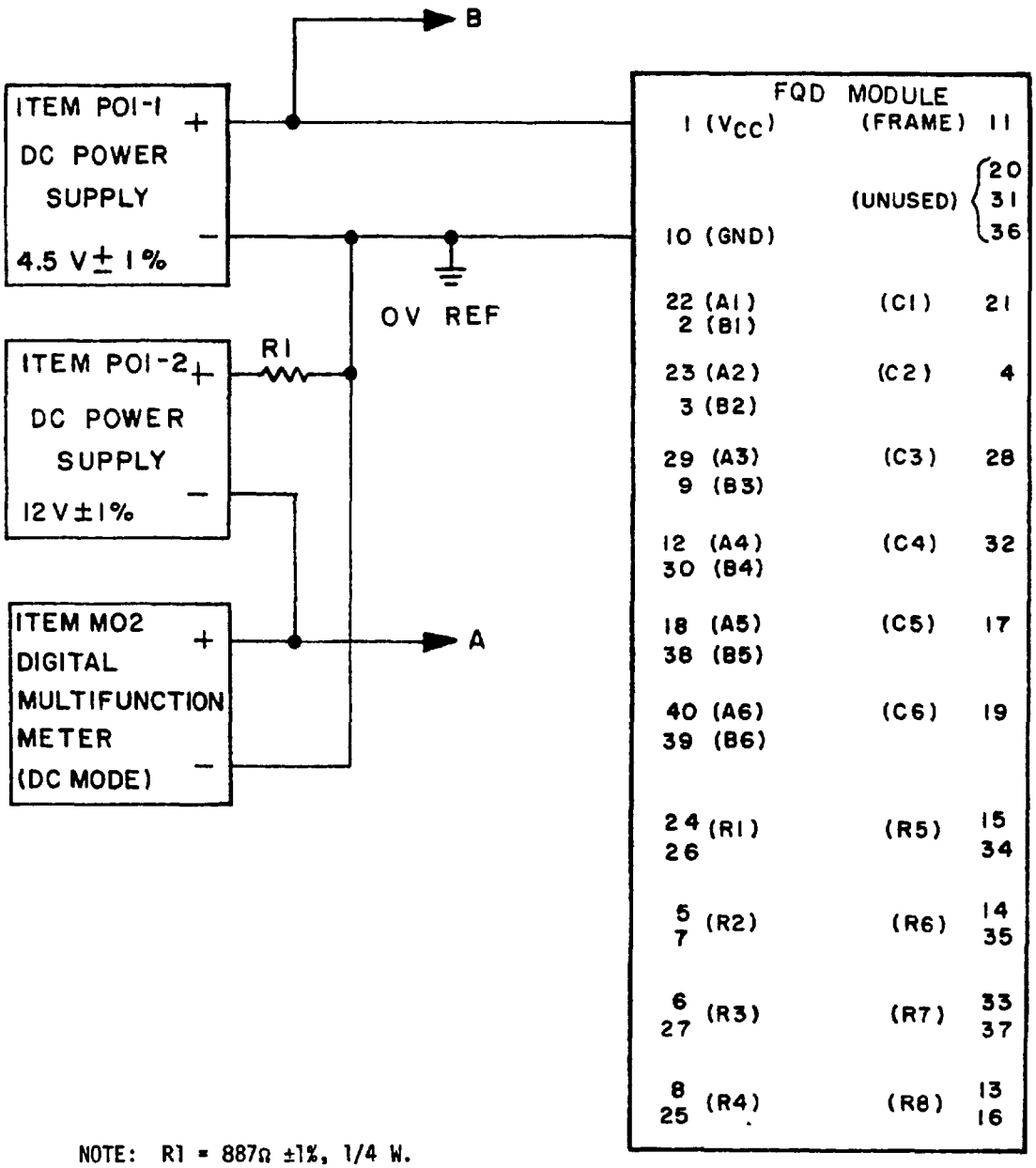
TABLE V. Input clamp diode voltage test.

| Test condition | | Parameter | Connect | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|
| | | | A to pin | B to pin | Min | Max | | |
| C1 | A1 | Input clamp diode voltage | 22 | 2 | -0.4 | -1.5 | V | 8 |
| C2 | B1 | " | 2 | 22 | " | " | " | " |
| C3 | A2 | " | 23 | 3 | " | " | " | " |
| C4 | B2 | " | 3 | 23 | " | " | " | " |
| C5 | A3 | " | 29 | 9 | " | " | " | " |
| C6 | B3 | " | 9 | 29 | " | " | " | " |
| C7 | A4 | " | 12 | 30 | " | " | " | " |
| C8 | B4 | " | 30 | 12 | " | " | " | " |
| C9 | A5 | " | 38 | 18 | " | " | " | " |
| C10 | B5 | " | 18 | 38 | " | " | " | " |
| C11 | A6 | " | 40 | 39 | " | " | " | " |
| C12 | B6 | " | 39 | 40 | " | " | " | " |





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NOTE: R1 = 887Ω ±1%, 1/4 W.

* FIGURE 8. Input clamp diode voltage test circuit for table V.



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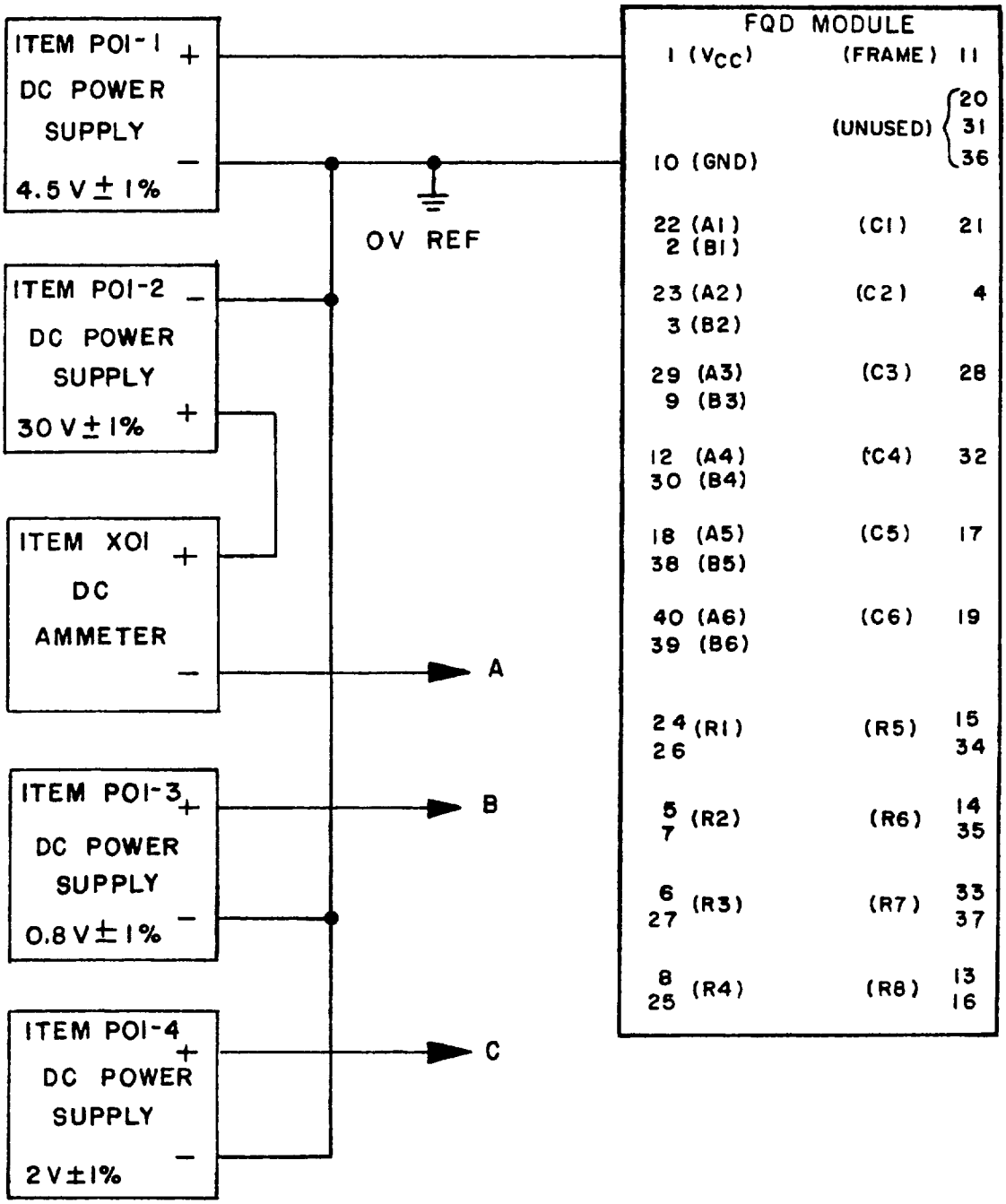
TABLE VI. Output leakage current test.

| Test condition | Parameter | Connect | | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | Min | Max | | |
| D1 | C1 Output leakage current | 21 | 22 | 2 | 0 | 300 | µA | 9 |
| D2 | C1 " | 21 | 2 | 22 | " | " | " | " |
| D3 | C2 " | 4 | 23 | 3 | " | " | " | " |
| D4 | C2 " | 4 | 3 | 23 | " | " | " | " |
| D5 | C3 " | 28 | 29 | 9 | " | " | " | " |
| D6 | C3 " | 28 | 9 | 29 | " | " | " | " |
| D7 | C4 " | 32 | 12 | 30 | " | " | " | " |
| D8 | C4 " | 32 | 30 | 12 | " | " | " | " |
| D9 | C5 " | 17 | 18 | 38 | " | " | " | " |
| D10 | C5 " | 17 | 38 | 18 | " | " | " | " |
| D11 | C6 " | 19 | 40 | 39 | " | " | " | " |
| D12 | C6 " | 19 | 39 | 40 | " | " | " | " |





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CAUTION:

Do not isolate the chassis of item XOI from earth ground.

NOTE: Current limit item POI-2 to 50 mA.

FIGURE 9. Output leakage current test circuit for table VI.



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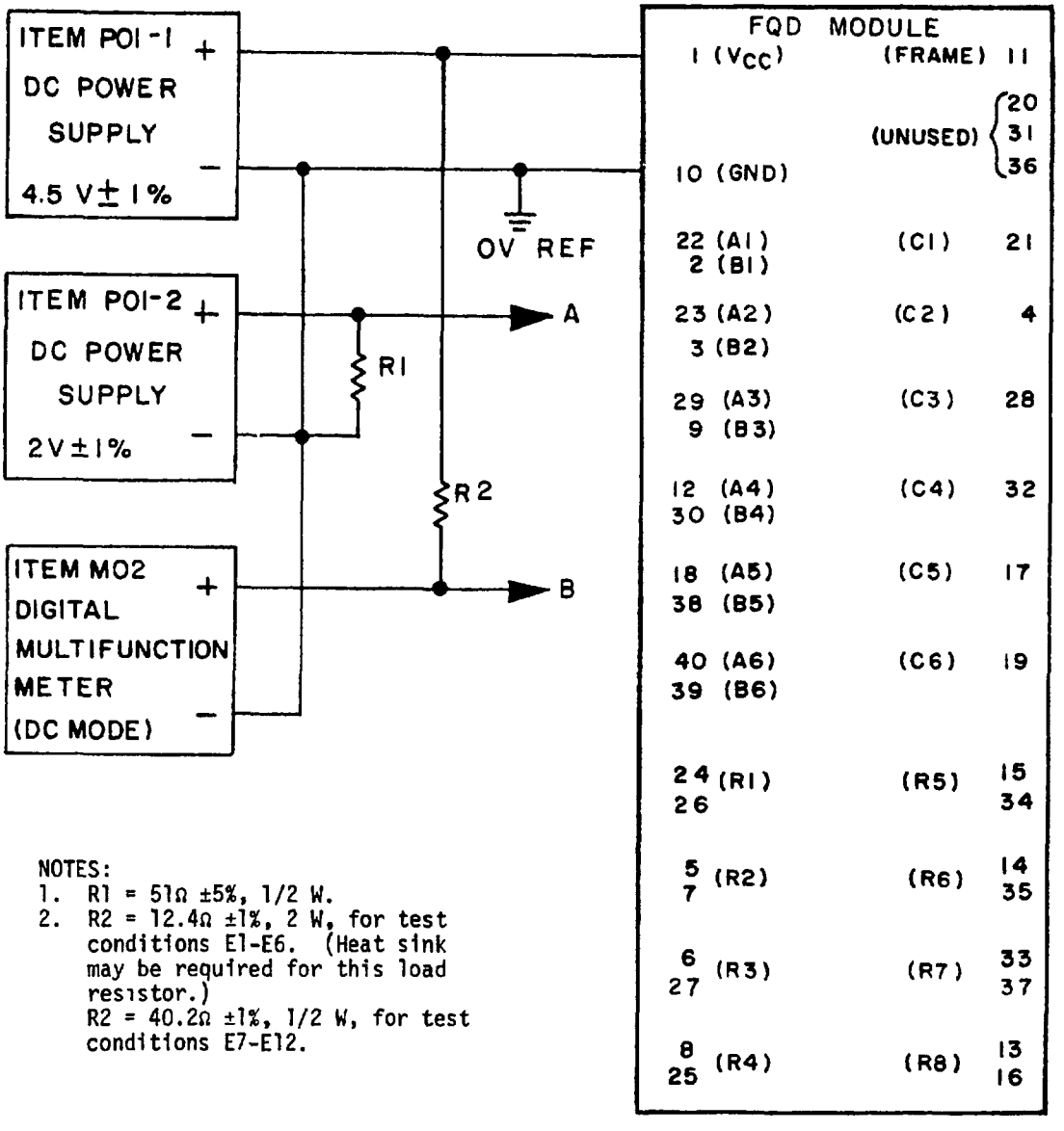
*TABLE VII. Output voltage (low-level) test.

| Test condition | Parameter | Connect | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | Min | Max | | |
| E1 | C1 Output voltage (low level), $I_{OL}$ = 300 mA | 2,22 | 21 | 0 | 800 | mV | 10 |
| E2 | C2 " | 3,23 | 4 | " | " | " | " |
| E3 | C3 " | 9,29 | 28 | " | " | " | " |
| E4 | C4 " | 12,30 | 32 | " | " | " | " |
| E5 | C5 " | 18,38 | 17 | " | " | " | " |
| E6 | C6 " | 39,40 | 19 | " | " | " | " |
| E7 | C1 Output voltage (low level), $I_{OL}$ = 100 mA | 2,22 | 21 | " | 500 | " | " |
| E8 | C2 " | 3,23 | 4 | " | " | " | " |
| E9 | C3 " | 9,29 | 28 | " | " | " | " |
| E10 | C4 " | 12,30 | 32 | " | " | " | " |
| E11 | C5 " | 18,38 | 17 | " | " | " | " |
| E12 | C6 " | 39,40 | 19 | " | " | " | " |





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NOTES:
1. R1 = 51Ω ±5%, 1/2 W.
2. R2 = 12.4Ω ±1%, 2 W, for test conditions E1-E6. (Heat sink may be required for this load resistor.)
   R2 = 40.2Ω ±1%, 1/2 W, for test conditions E7-E12.

* FIGURE 10. Output voltage (low-level) test circuit for table VII.



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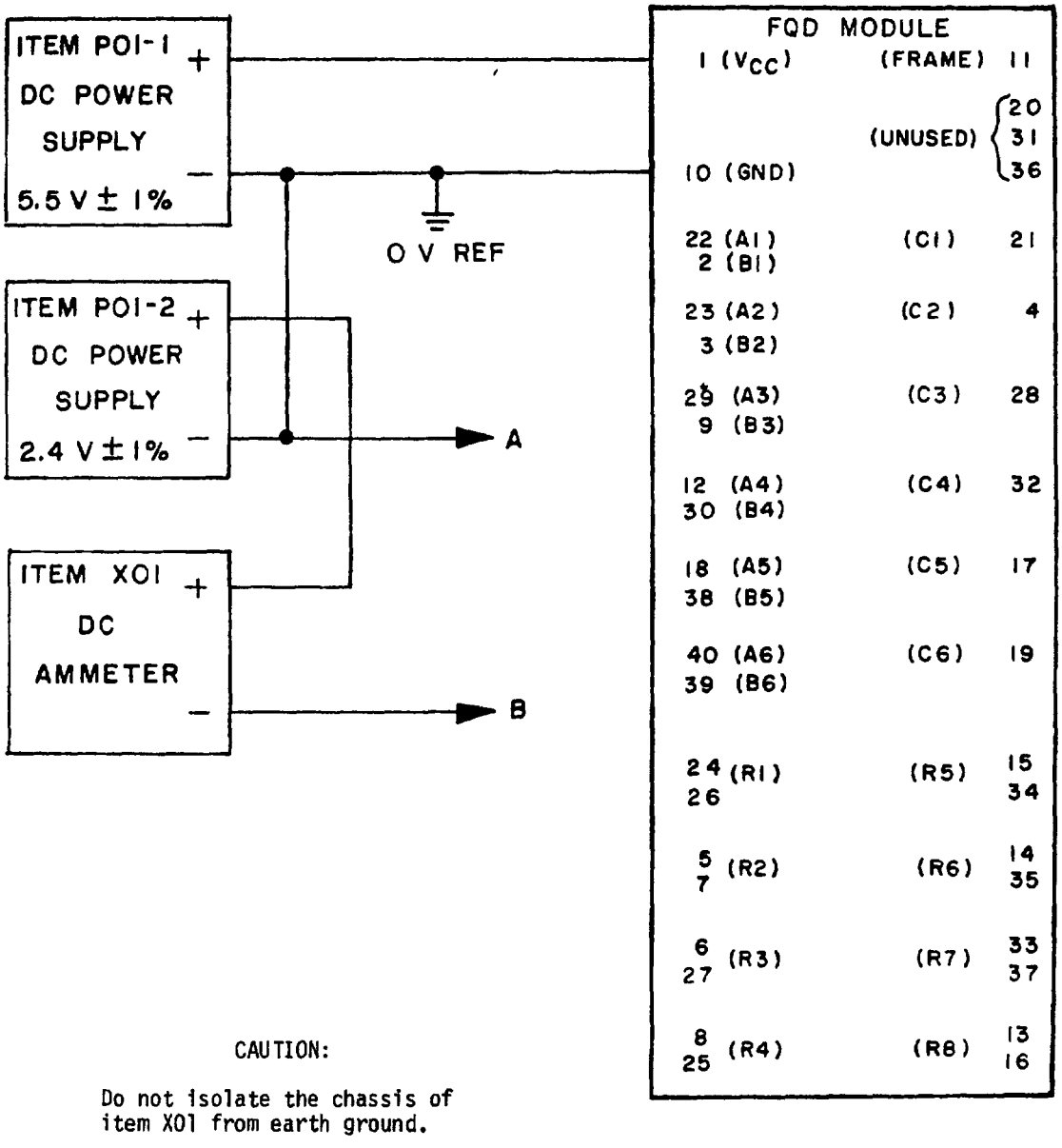
TABLE VIII. Input current (high-level) test.

| Test condition | Parameter | | Connect | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|
| | | | A to pin | B to pin | Min | Max | | |
| F1 | A1 | Input current, high level | 2,3,9,12,18,23, 29,30,38,39,40 | 22 | 0 | 40 | µA | 11 |
| F2 | B1 | " | 3,9,12,18,22, 23,29,30,38, 39,40 | 2 | " | " | " | " |
| F3 | A2 | " | 2,3,9,12,18,22, 29,30,38,39,40 | 23 | " | " | " | " |
| F4 | B2 | " | 2,9,12,18,22, 23,29,30,38,39, 40 | 3 | " | " | " | " |
| F5 | A3 | " | 2,3,9,12,18,22, 23,30,38,39,40 | 29 | " | " | " | " |
| F6 | B3 | " | 2,3,12,18,22, 23,29,30,38,39, 40 | 9 | " | " | " | " |
| F7 | A4 | " | 2,3,9,18,22,23, 29,30,38,39,40 | 12 | " | " | " | " |
| F8 | B4 | " | 2,3,9,12,18,22, 23,29,38,39,40 | 30 | " | " | " | " |
| F9 | A5 | " | 2,3,9,12,22,23, 29,30,38,39,40 | 18 | " | " | " | " |
| F10 | B5 | " | 2,3,9,12,18,22, 23,29,30,39,40 | 38 | " | " | " | " |
| F11 | A6 | " | 2,3,9,12,18,22, 23,29,30,38,39 | 40 | " | " | " | " |
| F12 | B6 | " | 2,3,9,12,18,22, 23,29,30,38,40 | 39 | " | " | " | " |





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CAUTION:

Do not isolate the chassis of item XO1 from earth ground.

FIGURE 11. Input current (high-level) test circuit for table VIII.



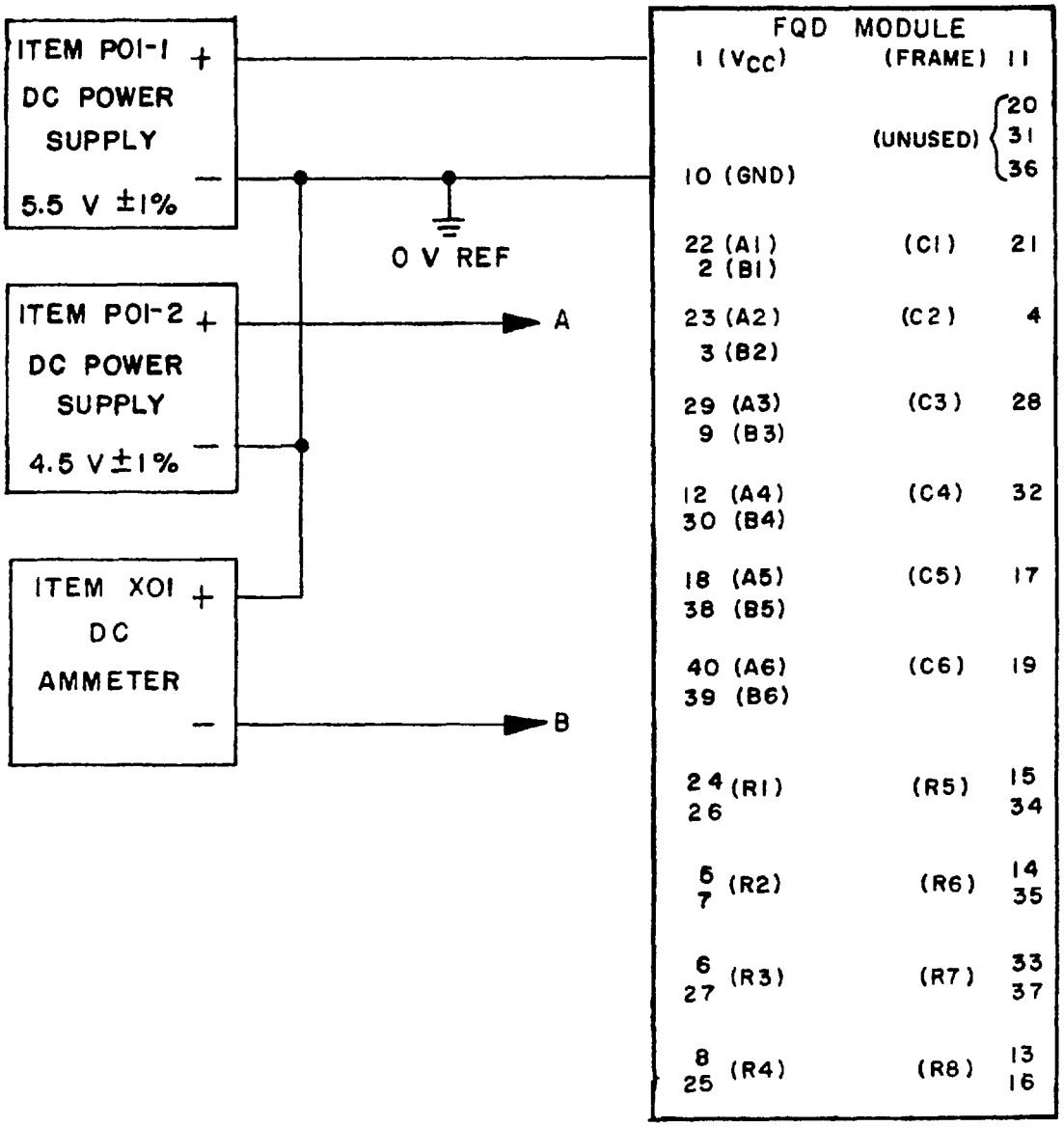
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TABLE IX. Input current (low-level) test.

| Test condition | Parameter | Connect | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | Min | Max | | |
| G1 | A1 Input current, low level | 2 | 22 | -0.4 | -1.73 | mA | 12 |
| G2 | B1 " | 22 | 2 | " | " | " | " |
| G3 | A2 " | 3 | 23 | " | " | " | " |
| G4 | B2 " | 23 | 3 | " | " | " | " |
| G5 | A3 " | 9 | 29 | " | " | " | " |
| G6 | B3 " | 29 | 9 | " | " | " | " |
| G7 | A4 " | 30 | 12 | " | " | " | " |
| G8 | B4 " | 12 | 30 | " | " | " | " |
| G9 | A5 " | 38 | 18 | " | " | " | " |
| G10 | B5 " | 18 | 38 | " | " | " | " |
| G11 | A6 " | 39 | 40 | " | " | " | " |
| G12 | B6 " | 40 | 31 | " | " | " | " |



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CAUTION:

Do not isolate the chassis of item X01 from earth ground.

FIGURE 12. Input current (low-level) test circuit for table IX.



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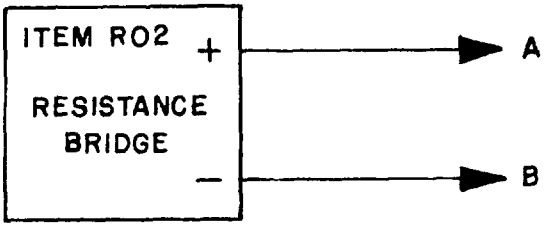
TABLE X. Resistance test.

| Test condition | Parameter | Connect | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | Min | Max | | |
| H1 | R1 Resistance | 24 | 26 | 950 | 1,050 | Ω | 13 |
| H2 | R2 " | 5 | 7 | " | " | " | " |
| H3 | R3 " | 6 | 27 | " | " | " | " |
| H4 | R4 " | 8 | 25 | " | " | " | " |
| H5 | R5 " | 15 | 34 | " | " | " | " |
| H6 | R6 " | 14 | 35 | " | " | " | " |
| H7 | R7 " | 33 | 37 | " | " | " | " |
| H8 | R8 " | 13 | 16 | " | " | " | " |





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| FQD MODULE | | |
|---|---|---|
| 1 ($V_{CC}$) | (FRAME) | 11 |
| 10 (GND) | (UNUSED) | 20, 31, 36 |
| 22 (A1), 2 (B1) | (C1) | 21 |
| 23 (A2), 3 (B2) | (C2) | 4 |
| 29 (A3), 9 (B3) | (C3) | 28 |
| 12 (A4), 30 (B4) | (C4) | 32 |
| 18 (A5), 38 (B5) | (C5) | 17 |
| 40 (A6), 39 (B6) | (C6) | 19 |
| 24, 26 (R1) | (R5) | 15, 34 |
| 5, 7 (R2) | (R6) | 14, 35 |
| 6, 27 (R3) | (R7) | 33, 37 |
| 8, 25 (R4) | (R8) | 13, 16 |

FIGURE 13. Resistance test circuit for table X.



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TABLE XI. Power supply current (low-level and high-level outputs) test.

| Test condition | Parameter | Connect | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | Min | Max | | |
| J1 | Power supply current, low level outputs | C | Open | 1 | 213 | mA | 14 |
| J2 | Power supply current, high level outputs | Open | C | 1 | 42 | mA | 14 |

TABLE XII. Transition time ($t_{THL}$) test.

| Test condition | Parameter | Connect | | | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | D to pin | Min | Max | | |
| K1 | C1 Transition time ($t_{THL}$), $C_L$ = 50 pF | 22,23,29, 12,18,40 | 2 | Open | 21 | 2 | 14 | ns | 15,16 |
| K2 | C2 " | " | 3 | " | 4 | " | " | " | " |
| K3 | C3 " | " | 9 | " | 28 | " | " | " | " |
| K4 | C4 " | " | 30 | " | 32 | " | " | " | " |
| K5 | C5 " | " | 38 | " | 17 | " | " | " | " |
| K6 | C6 " | " | 39 | " | 19 | " | " | " | " |





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TABLE XIII. Propagation delay time ($t_{PLH}$ and $t_{PHL}$) test.

| Test condition | Parameter | | Connect | | | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | A to pin | B to pin | C to pin | D to pin | Min | Max | | |
| L1 | A1 to C1 | Propagation delay time ($t_{PLH}$), $C_L$ = 50 pF | 2,3,9,12,18, 23,29,30,38, 39,40 | 22 | B | 21 | 3 | 36 | ns | 15,16 |
| L2 | B1 to C1 | " | 3,9,12,18, 22,23,29,30, 38,39,40 | 2 | " | 21 | " | " | " | " |
| L3 | A2 to C2 | " | 2,3,9,12,18, 22,29,30,38, 39,40 | 23 | " | 4 | " | " | " | " |
| L4 | B2 to C2 | " | 2,9,12,18, 22,23,29,30, 38,39,40 | 3 | " | 4 | " | " | " | " |
| L5 | A3 to C3 | " | 2,3,9,12,18, 22,23,30,38, 39,40 | 29 | " | 28 | " | " | " | " |
| L6 | B3 to C3 | " | 2,3,12,18, 22,23,29,30, 38,39,40 | 9 | " | 28 | " | " | " | " |
| L7 | A4 to C4 | " | 2,3,9,18, 22,23,29,30, 38,39,40 | 12 | " | 32 | " | " | " | " |
| L8 | B4 to C4 | " | 2,3,9,12,18, 22,23,29,38, 39,40 | 30 | " | 32 | " | " | " | " |
| L9 | A5 to C5 | " | 2,3,9,12,22, 23,29,30,38, 39,40 | 18 | " | 17 | " | " | " | " |
| L10 | B5 to C5 | " | 2,3,9,12,18, 22,23,29,30, 39,40 | 38 | " | 17 | " | " | " | " |
| L11 | A6 to C6 | " | 2,3,9,12,18, 22,23,29,30, 38,39 | 40 | " | 19 | " | " | " | " |



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TABLE XIII. Propagation delay time ($t_{PLH}$ and $t_{PHL}$) test - Continued.

| Test condition | Parameter | | Connect | | | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | A to pin | B to pin | C to pin | D to pin | Min | Max | | |
| L12 | B6 to C6 | Propagation delay time ($t_{PLH}$), $C_L$ = 50 pF | 2,3,9,12,18, 22,23,29,30, 38,40 | 39 | B | 19 | 3 | 36 | ns | 15,16 |
| M1 | A1 to C1 | Propagation delay time ($t_{PHL}$), $C_L$ = 50 pF | 2,3,9,12,18, 23,29,30,38, 39,40 | 22 | " | 21 | " | " | " | " |
| M2 | B1 to C1 | " | 3,9,12,18, 22,23,29,30, 38,39,40 | 2 | " | 21 | " | " | " | " |
| M3 | A2 to C2 | " | 2,3,9,12,18, 22,29,30,38, 39,40 | 23 | " | 4 | " | " | " | " |
| M4 | B2 to C2 | " | 2,9,12,18, 22,23,29,30, 38,39,40 | 3 | " | 4 | " | " | " | " |
| M5 | A3 to C3 | " | 2,3,9,12,18, 22,23,30,38, 39,40 | 29 | " | 28 | " | " | " | " |
| M6 | B3 to C3 | " | 2,3,12,18, 22,23,29,30, 38,39,40 | 9 | " | 28 | " | " | " | " |
| M7 | A4 to C4 | " | 2,3,9,18, 22,23,29,30, 38,39,40 | 12 | " | 32 | " | " | " | " |
| M8 | B4 to C4 | " | 2,3,9,12,18, 22,23,29,38, 39,40 | 30 | " | 32 | " | " | " | " |
| M9 | A5 to C5 | " | 2,3,9,12,22, 23,29,30,38, 39,40 | 18 | " | 17 | " | " | " | " |
| M10 | B5 to C5 | " | 2,3,9,12,18, 22,23,29,30, 39,40 | 38 | " | 17 | " | " | " | " |





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TABLE XIII. Propagation delay time ($t_{PLH}$ and $t_{PHL}$) test - Continued.

| Test condition | Parameter | Connect | | | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | D to pin | Min | Max | | |
| M11 | A6 to C6 Propagation delay time ($t_{PHL}$), $C_L$ = 50 pF | 2,3,9,12,18, 22,23,29,30, 38,39 | 40 | B | 19 | 3 | 36 | ns | 15,16 |
| M12 | B6 to C6 Propagation delay time ($t_{PHL}$), $C_L$ = 50 pF | 2,3,9,12,18, 22,23,29,30, 38,40 | 39 | B | 19 | 3 | 36 | ns | 15,16 |





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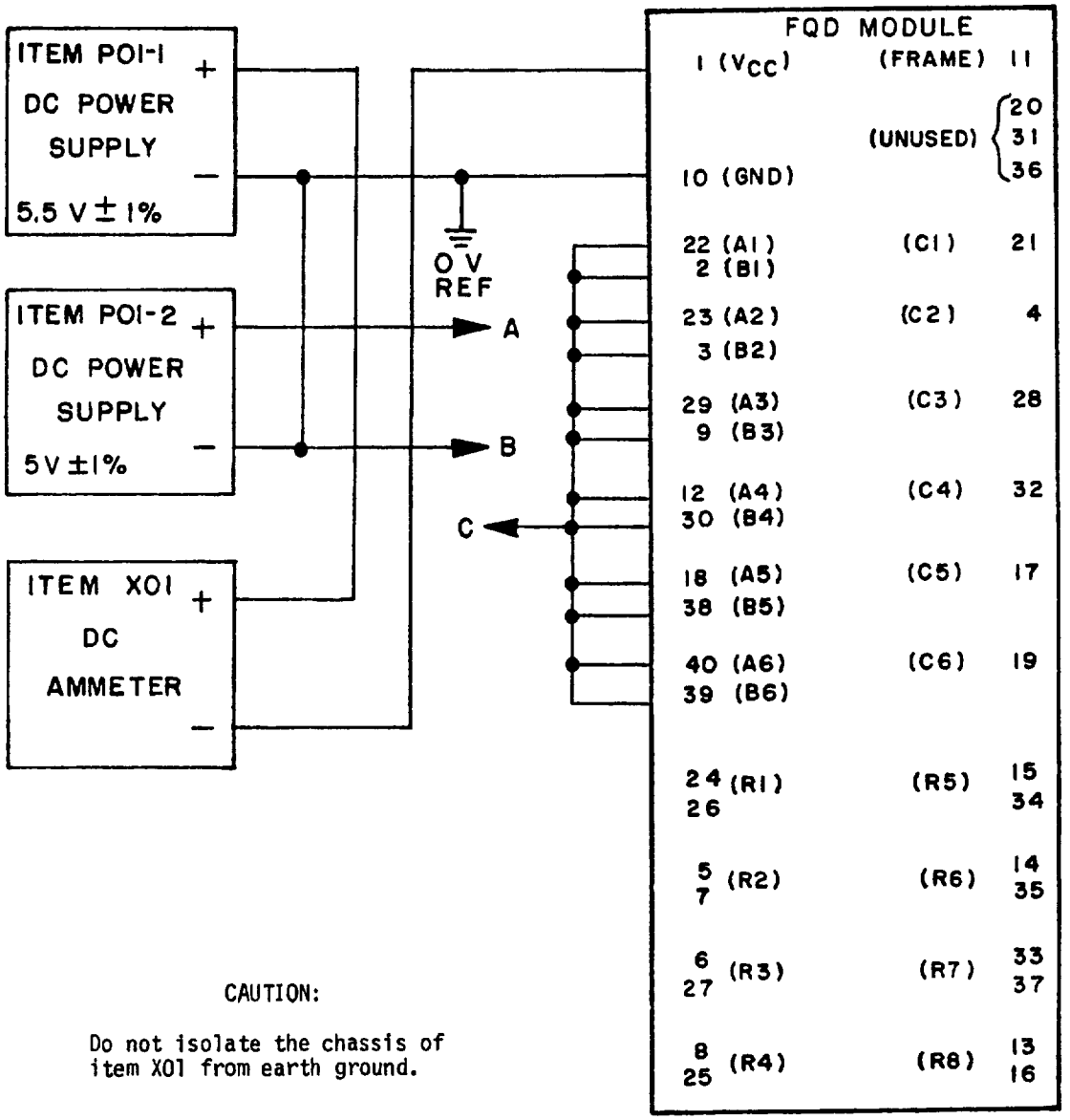
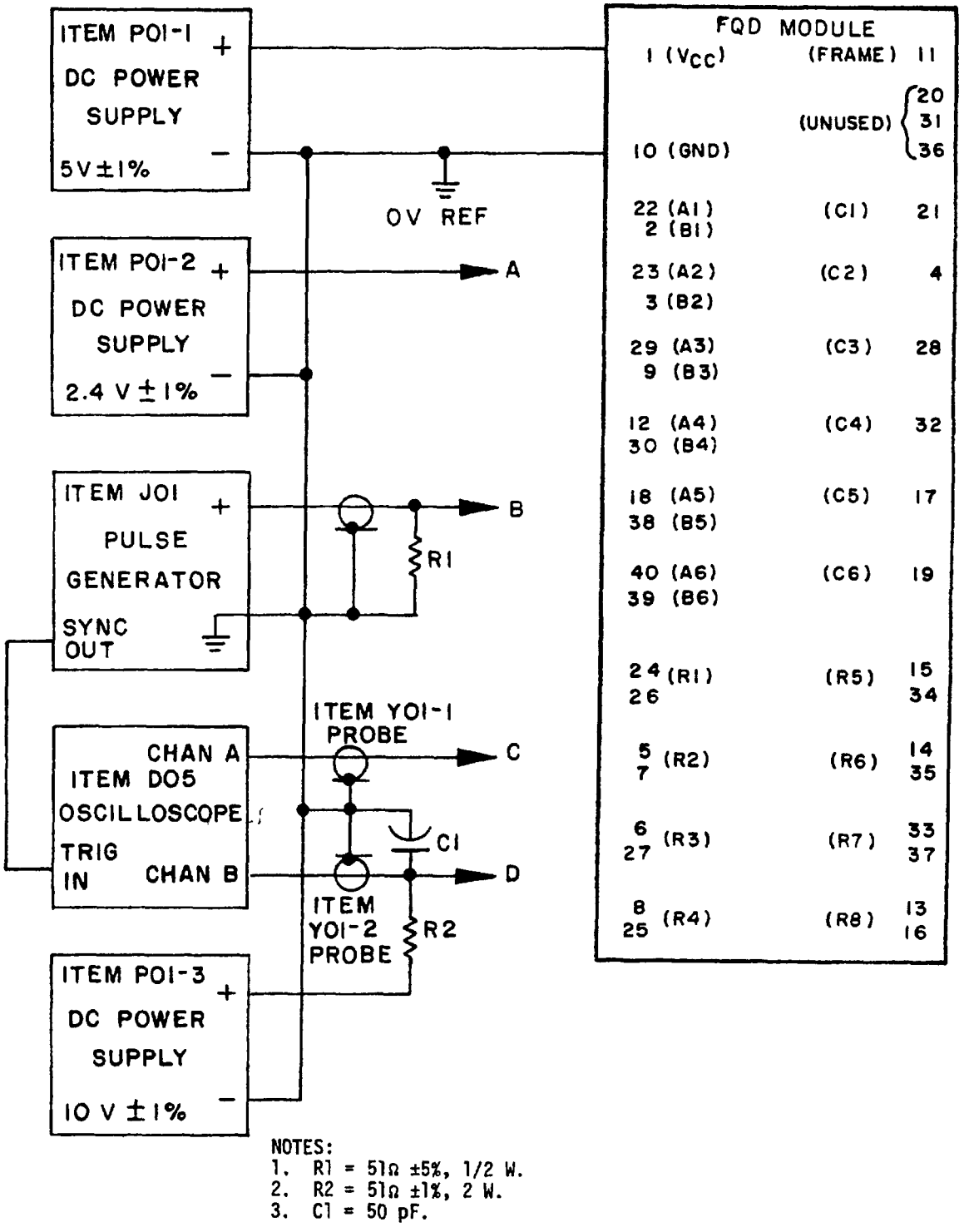



FIGURE 14. Power supply current (low-level and high-level outputs) test circuit for table XI.



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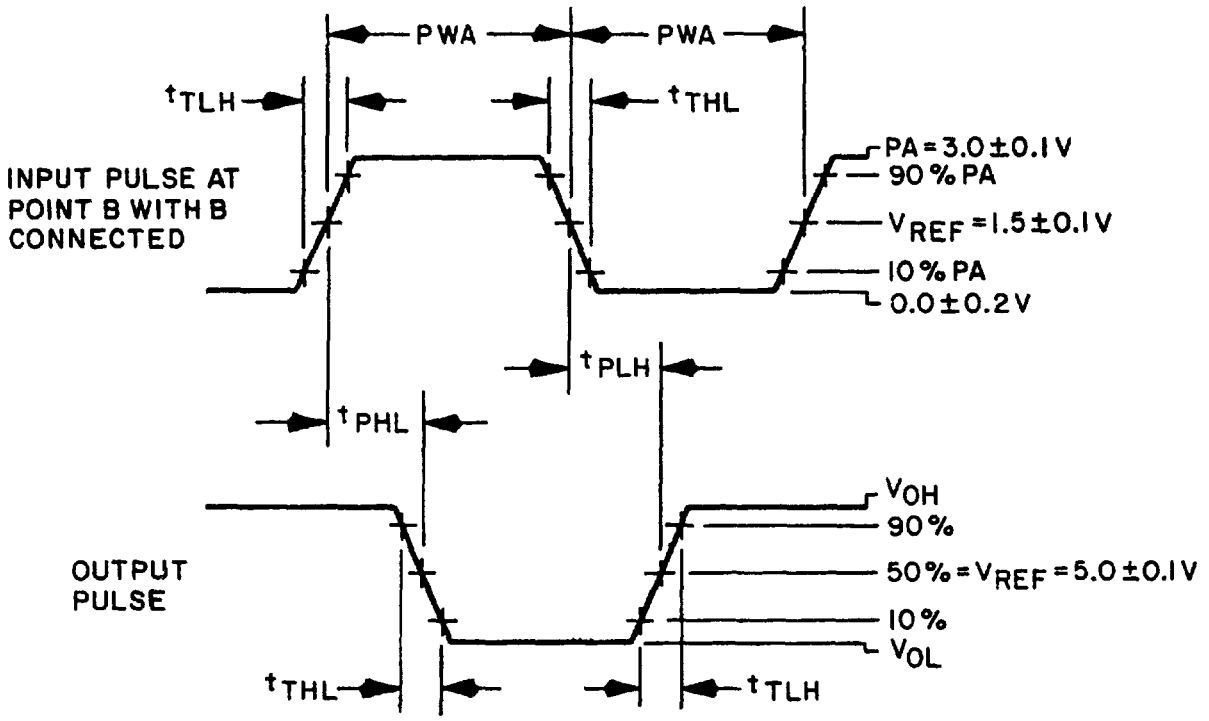



NOTES:
1. R1 = 51Ω ±5%, 1/2 W.
2. R2 = 51Ω ±1%, 2 W.
3. C1 = 50 pF.

FIGURE 15. Transition time ($t_{THL}$) and propagation delay time ($t_{PLH}$ and $t_{PHL}$) test circuit for tables XII and XIII.



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NOTES:
1. Input pulse parameters:
   PWA = 500 ±50 ns.
   $t_{TLH} = t_{THL} = 10 \pm 2$ ns.
2. All voltages are referenced to ground.
3. Before taking measurements, verify that the output high level is $\geq$ 9.5 V and the low level is $\leq$ 0.8 V.

* FIGURE 16. Transition time ($t_{THL}$) and propagation delay time ($t_{PLH}$ and $t_{PHL}$) test circuit waveforms for tables XII and XIII.



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TABLE XIV. Power supply transient amplitude test.

| Test condition | Parameter | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|
| | | Min | Max | | |
| N | Power supply transient amplitude | 0 | 400 | mV | 17,18 |

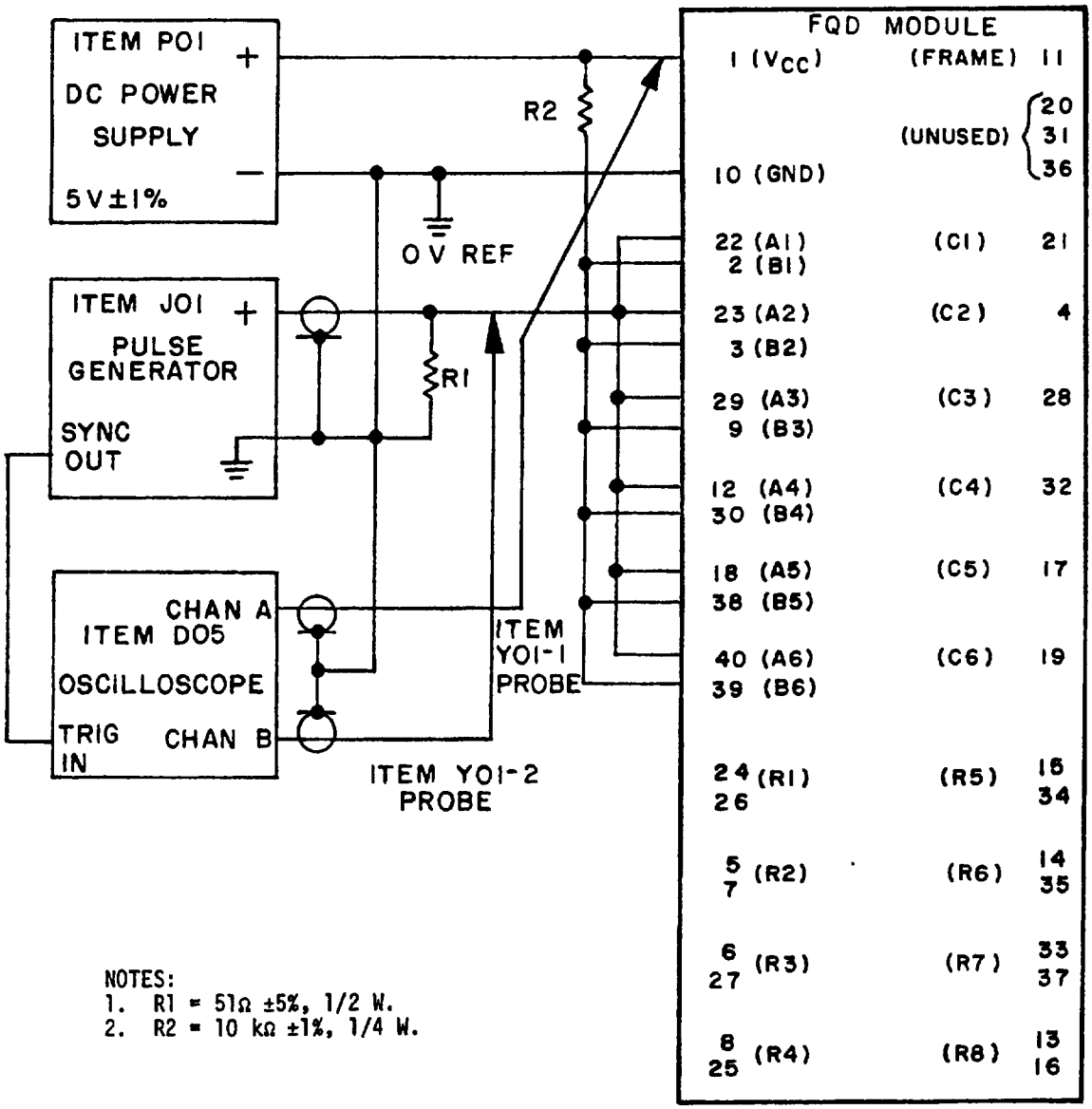
TABLE XV. Power supply filter capacitance test.

| Test condition | Parameter | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|
| | | Min | Max | | |
| P | Power supply filter capacitance | 3.3 | 6.1 | µF | 19 |





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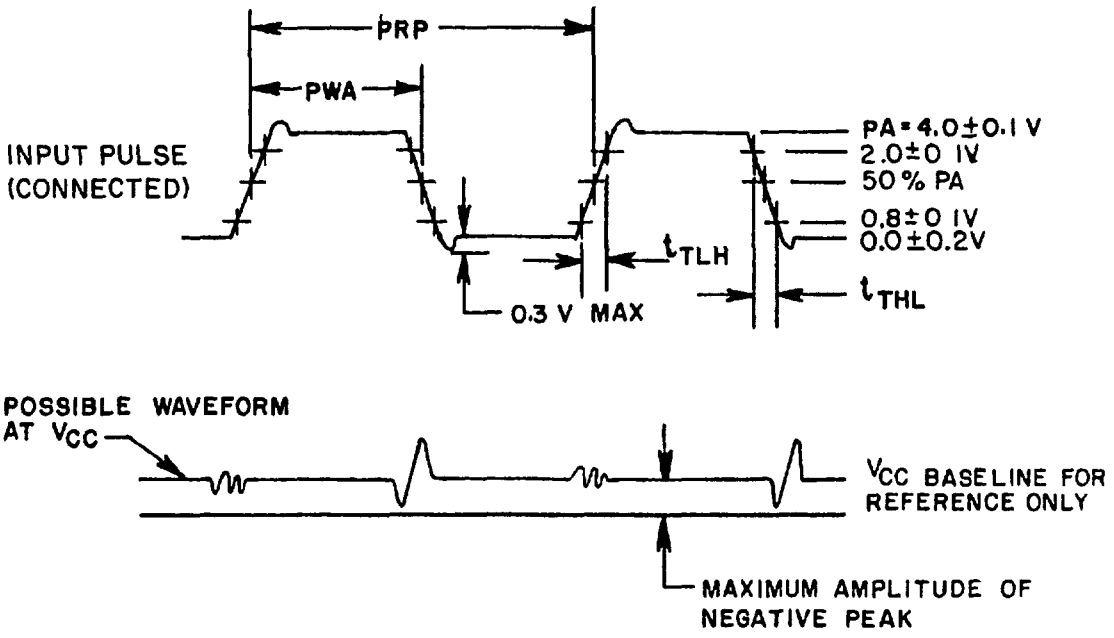



NOTES:
1. R1 = 51Ω ±5%, 1/2 W.
2. R2 = 10 kΩ ±1%, 1/4 W.

FIGURE 17. Power supply transient amplitude test circuit for table XIV.



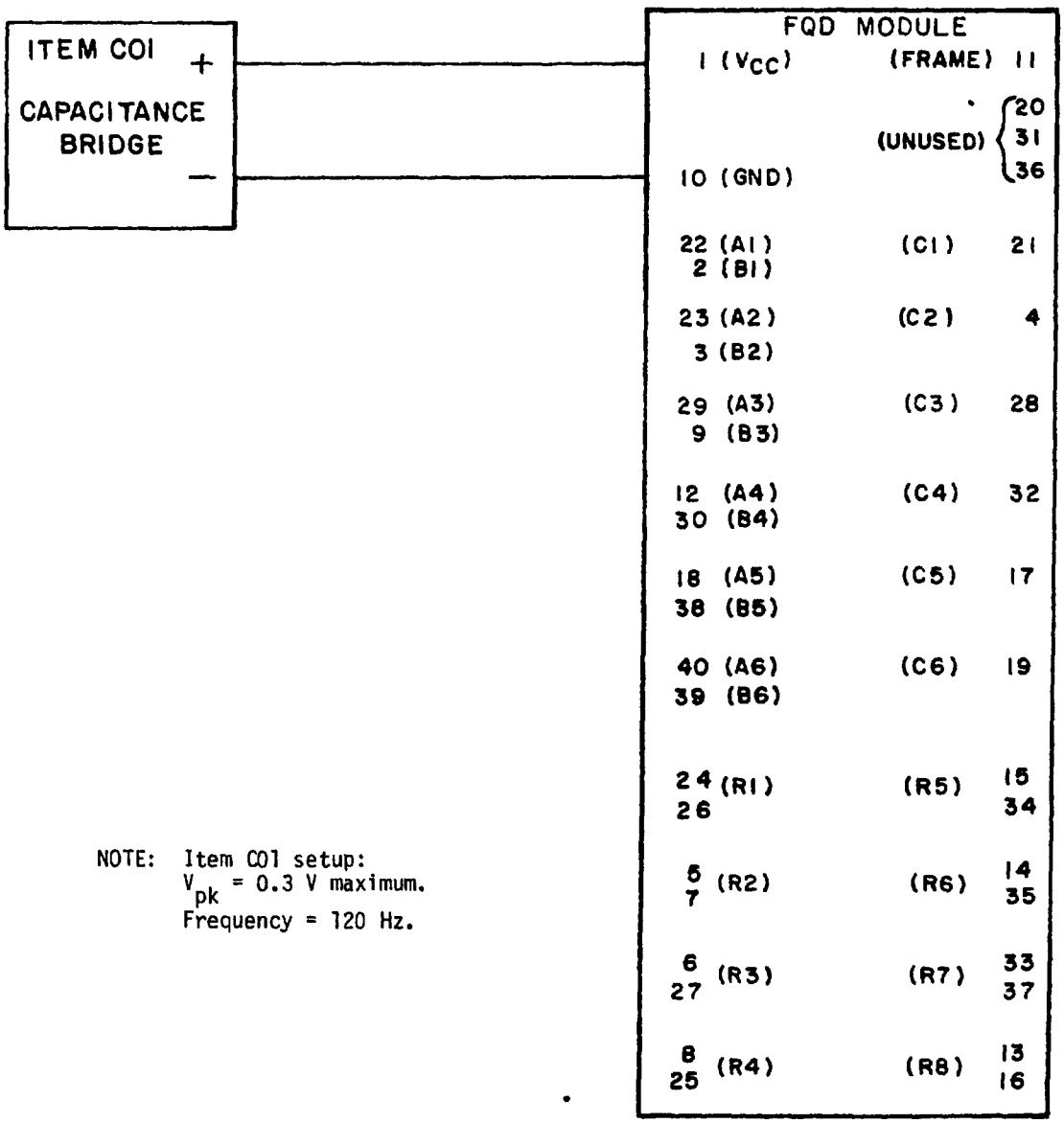
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:21Z
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NOTE: Input pulse parameters:
PWA = 2.5 ±.25 µs.
PRP = 5.0 ±.5 µs.
Vary $t_{TLH}$ from 5 ns to 25 ns and $t_{THL}$ from 3 ns to 10 ns and record the maximum value of transient amplitude.

FIGURE 18. Power supply transient amplitude test circuit waveforms for table XIV.


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NOTE: Item COI setup:
$V_{pk}$ = 0.3 V maximum.
Frequency = 120 Hz.

FIGURE 19. Power supply filter capacitance test circuit for table XV.



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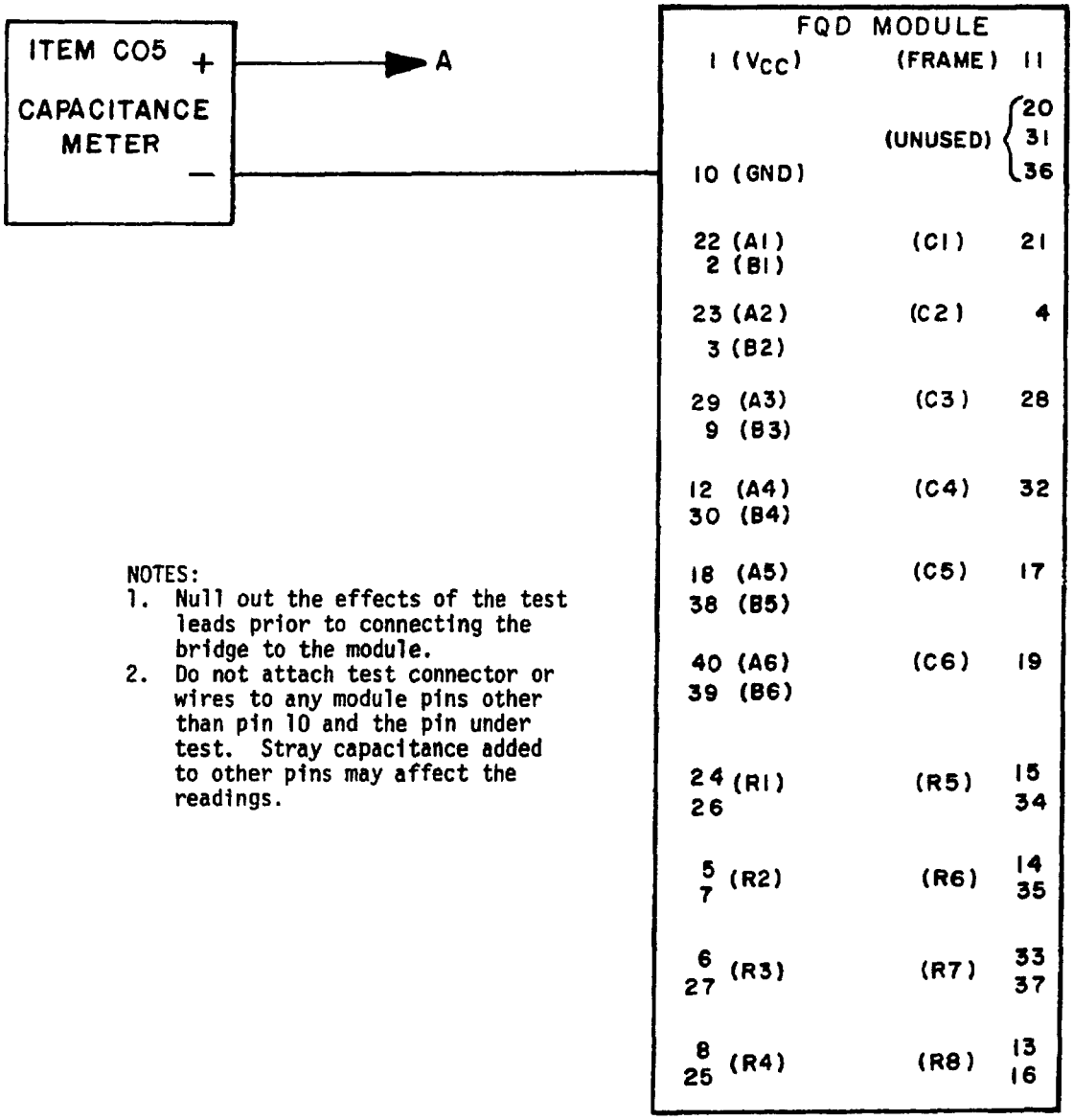
TABLE XVI. Input capacitance test.

| Test condition | Parameter | Connect A to pin | Initial limits 25°C ±5°C ambient Min | Initial limits 25°C ±5°C ambient Max | Unit | Test figure |
|---|---|---|---|---|---|---|
| Q1 | A1 Input capacitance | 22 | 2 | 15 | pF | 20 |
| Q2 | B1 " | 2 | " | " | " | " |
| Q3 | A2 " | 23 | " | " | " | " |
| Q4 | B2 " | 3 | " | " | " | " |
| Q5 | A3 " | 29 | " | " | " | " |
| Q6 | B3 " | 9 | " | " | " | " |
| Q7 | A4 " | 12 | " | " | " | " |
| Q8 | B4 " | 30 | " | " | " | " |
| Q9 | A5 " | 18 | " | " | " | " |
| Q10 | B5 " | 38 | " | " | " | " |
| Q11 | A6 " | 40 | " | " | " | " |
| Q12 | B6 " | 39 | " | " | " | " |





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NOTES:
1. Null out the effects of the test leads prior to connecting the bridge to the module.
2. Do not attach test connector or wires to any module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.

FIGURE 20. Input capacitance test circuit for table XVI.



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| MIL-M-28787/278A(EC) | Modules, Standard Electronic, Driver, Lamp/Relay, Digital, Key Code FQD |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR<br>☐ USER<br>☐ MANUFACTURER<br>☐ OTHER *(Specify)* _____ |
| b. ADDRESS *(Street, City, State, ZIP Code)* | |

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
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